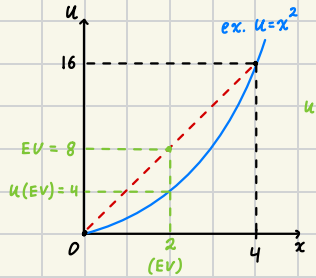
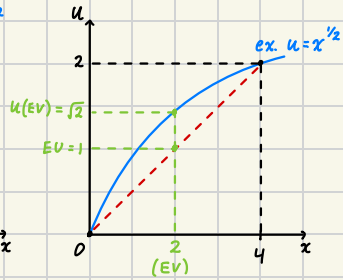
Recap: Attitudes towards risk

ex. coin flip → Heads \$4 ↳ Tails \$0 } prob = 0.5 for each

Risk Loving



Risk Aversion



Side Note

Experimental evidence

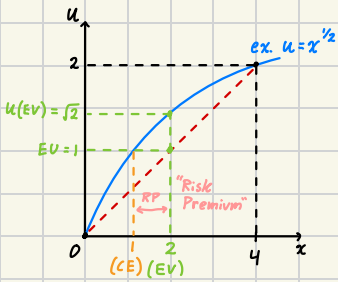
A risk averse decision maker prefers to get EV for sure instead of lottery

→ they would actually accept less than EV

→ what is the lowest amount of money that a risk averse person would accept?

⇒ certainty equivalent

Same example: coin flip → \$4, \$0



Certainty equivalent is the lowest amount this person would accept instead of lottery

→ amount that gives the same utility as the lottery

$x^{\frac{1}{2}} = 1$ → EU

$\Rightarrow x = 1$

Note: for someone who is risk averse, the certainty equivalent is lower than EV

the difference between CE and EV is called the "risk premium"

CE and RP depend both on lottery and utility function

more risk averse → CE↓ RP↑

Demand for Insurance

many people are willing to pay to avoid risk

→ insurance reduces risk by raising income in bad states (making a claim) and lowering income in good states (paying premium)

ex. you have a car worth \$10,000 and over the next year: 10% chance of accident
accident: value of car → \$2500
your preferences are $u = x^{1/2}$ (x is the value of the car)

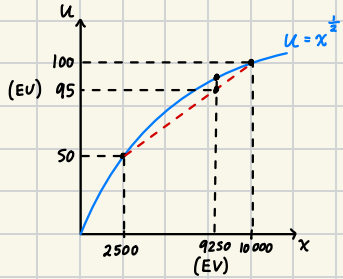
what is the expected value of your car?

$$EV = (0.9)(10000) + (0.1)(2500) = 9250$$

what is your expected utility?

$$EU = (0.9)(10000)^{\frac{1}{2}} + (0.1)(2500)^{\frac{1}{2}} = 95$$

what if an insurance company offers you a policy that fully covers loss (value of car is always \$10 000)



→ would you pay \$750 for this policy?

yes, this would give you \$9250 (EV) with certainty

what is the maximum amount you would pay?

→ need to find certainty equivalent

remember $EU = 95$ → need a value of x that gives $u = 95$

$x^{\frac{1}{2}} = 95 \Rightarrow x = 9025 = CE$

so max willingness to pay for full insurance is $10000 - 9025 = \$975$

Cooperatives & Risk Sharing

what if you don't have access to insurance in your community? → cooperatives

cooperatives are a way of "sharing" risk
- work when risks are independent

ex. two farmers face an independent risk of crop failure, $u = x^{\frac{1}{2}}$

→ if crop succeeds $\$100$ (prob $\frac{1}{2}$)
if crop fails $\$0$ (prob $\frac{1}{2}$)

alone $EU = (\frac{1}{2})(100)^{\frac{1}{2}} + (\frac{1}{2})(0)^{\frac{1}{2}} = 5$

because risks are independent,

		farmer B	
		$100	$0
farmer A	$100	1/4	1/4
	$0	1/4	1/4

← joint probabilities

suppose they form a cooperative: if one succeeds and the other fails, split revenue

what is expected utility now?

$EU = (\frac{1}{4})(100)^{\frac{1}{2}} + (\frac{1}{4})(50)^{\frac{1}{2}} + (\frac{1}{4})(50)^{\frac{1}{2}} + (\frac{1}{4})(0)^{\frac{1}{2}} = 6.04$ (both are better off)

Note: risks must be independent (can't insure against aggregate risk)